

SMART HOME AUTOMATION WITH REAL - TIME MONITORING AND ACTUATION

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A report for the 23EC353-Mini Project-II submitted to the faculty of

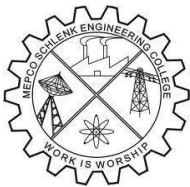
ELECTRONICS AND COMMUNICATION ENGINEERING

for partial fulfillment of the award of the degree of

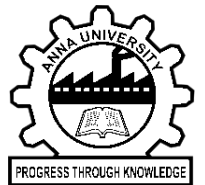
BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING



**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**



MEPCO SCHLENK ENGINEERING COLLEGE, SIVAKASI

(An Autonomous Institution affiliated to Anna University Chennai)

May 2026

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CERTIFICATE

This is to certify that it is the Bonafide work done by Mr. NAVEEN PRAKASH. R (9517202403103), Mr. SIDDESHWARAN. V.P (9517202403154) for the 23EC353 - Mini Project - II work entitled Smart Home Automation With Real-Time Monitoring and Actuation at Department of Electronics and Communication Engineering, Mepco Schlenk Engineering College, Sivakasi during the year 2025 – 2026.

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Internal Examiner - I

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Internal Examiner - II

ABSTRACT

The rapid advancement of home automation technologies has played a significant role in improving convenience, safety and energy efficiency in modern living environments. With increasing demand for intelligent systems, there is a growing need for solutions that can automatically monitor surroundings and respond to real-time conditions without constant human intervention. This project presents the design and implementation of a multi-functional smart home automation system that integrates six essential modules, including motion detection, gas leakage detection, ambient light sensing, temperature monitoring, appliance control and security-based operation.

The system is developed using a microcontroller-based architecture, where sensors such as PIR sensors, MQ gas sensor, LDR and thermistor continuously monitor environmental parameters. These sensors provide real-time data to the microcontroller, enabling it to make decisions based on predefined conditions. Depending on the inputs received, the system performs automated actions such as controlling a fan using PWM-based speed regulation, switching lighting systems, activating alert mechanisms like buzzers and operating a servo-based door system.

A key feature of the proposed system is the inclusion of a security mode, which alters the behaviour of certain modules to enhance safety during critical conditions. For example, intrusion detection and restricted door access ensure better security, while hazard detection systems remain active at all times. The use of a motor driver module further enables efficient control of actuators such as DC motors.

The proposed system emphasizes low-cost implementation, simplicity and ease of integration, making it suitable for practical residential applications. Its modular structure allows individual components to be maintained or upgraded without affecting the entire system. Furthermore, the design provides a strong foundation for future enhancements, such as integration with Internet of Things (IoT) platforms for remote monitoring and control via smartphones or web interfaces.

Overall, this project demonstrates an effective, scalable and user-friendly approach to smart home automation, contributing to improved comfort, enhanced safety and better energy management in everyday living spaces.

ACKNOWLEDGEMENT

First and foremost, we express our wholehearted gratitude to the **Almighty** and **our parents** for being with us throughout extending moral support and helping us in completing this project work.

We wish to express our sincere thanks to **Dr.S. Arivazhagan M.E., Ph.D., F.I.E.T.E., M.I.S.T.E.**, honourable principal of our institution for offering us to take up the project in the college and extending all the facilities for the completion of the same.

Our heartfelt acknowledgement goes to the professor and Head of the Department of Electronics and Communication Engineering, **Dr.R. Shantha Selva Kumari, B.E., M.S., Ph.D., F.I.E.T.E., M.I.S.T.E.**, for her valuable and consistent encouragement for carrying out the project.

We express our deepest gratitude to our guide **Dr.W. Sylvia Lilly Jebarani, M.E., Ph.D.**, Professor of the Department of Electronics and Communication Engineering, for rendering excellent guidance and for extremely kind and approachable.

We hereby acknowledge the tireless efforts of all the staff members, reviewers and technicians of Electronics and Communication Department whose help was instrumental in the completion of our project. We extend our sincere thanks to our friends for playing a vital role in completing this work, by supporting us in all our endeavours.

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CHAPTER 1

INTRODUCTION

1.1 PROJECT BACKGROUND:

1.1.1 INTRODUCTION TO SMART HOME AUTOMATION:

In recent years, the rapid advancement of embedded systems and sensor technologies has led to the widespread development of smart home automation systems. These systems aim to improve quality of life by enhancing convenience, safety and energy efficiency in residential environments. Traditional homes rely on manual operation of electrical appliances, often resulting in inefficient energy usage and lack of real-time monitoring.

One major challenge in conventional home setups is the absence of integrated systems that can monitor environmental conditions and respond to hazards such as gas leakage, fire or unauthorized intrusion. This creates a need for a reliable and cost-effective automation system that ensures safety and efficient resource utilization.

To address these challenges, this project proposes a multi-module smart home automation system based on a microcontroller. The system integrates sensors such as PIR for motion detection, MQ for smoke detection, LDR for light sensing and a thermistor for temperature monitoring. These components provide real-time data to the microcontroller, which processes inputs and makes control decisions.

The system includes features like automated fan control based on temperature and occupancy, smoke detection with alerts, intrusion detection and automatic lighting. PWM control enables efficient motor speed regulation, while the modular design ensures flexibility and ease of expansion.

This system can be used in homes, offices and security-sensitive environments, with potential extension to IoT-based remote monitoring and control.

In conclusion, the proposed smart home automation system offers a practical, low-cost and scalable solution that enhances comfort, improves safety and promotes efficient energy management.

1.1.2 ADVANTAGES OF SMART HOME AUTOMATION:

- **Improved Convenience** - The system automates household operations such as lighting and appliance control, reducing the need for manual intervention.
- **Enhanced Safety** - Integration of gas sensors and motion detectors helps in detecting hazards like gas leakage and unauthorized movement, improving home security.
- **Energy Efficiency** - Devices are operated only when required (e.g., lights controlled using LDR), which minimizes unnecessary power consumption. Low maintenance and fewer electronic components make the system reliable and easy to maintain.
- **Low-Cost Implementation** - The use of commonly available sensors and a microcontroller makes the system affordable compared to commercial automation solutions.
- **Modular Design** - Each module (PIR, MQ sensor, LDR, thermistor, motor control) operates independently, allowing easy maintenance and future expansion.

1.1.3 APPLICATIONS OF SMART HOME AUTOMATION:

- **Residential Homes** - Smart home automation systems are widely used in houses to control lighting, fans and appliances automatically, improving comfort and reducing manual effort.
- **Office and Workplace Automation** - Automation improves efficiency in offices by managing lighting, temperature and security system.
- **Fire and Gas Safety Systems** - Smoke and gas detection modules can alert users and trigger safety actions, reducing the risk of accidents.
- **Home Security Systems** - Integration of motion sensors, alarms and security modes enhances protection against intrusions and unauthorized access..
- **IoT-Based Remote Monitoring Systems** - With IoT integration, users can monitor and control their home appliances remotely using smartphones or web applications..

1.2 OBJECTIVE OF THE PROJECT:

- To design and develop a multi-module smart home automation system using a microcontroller.
- To implement automatic control of household appliances based on sensor inputs.

CHAPTER 2

SYSTEM ARCHITECTURE

2.1 SYSTEM OVERVIEW:

The proposed system is a multi-module smart home automation system designed to enhance safety, convenience and energy efficiency in residential environments. It is built around a microcontroller-based architecture, where the Arduino Uno acts as the central control unit, coordinating the operation of all sensors and actuators.

The system consists of six functional modules, namely security mode control, temperature-based fan control, smoke detection, door control, intruder alert and automatic lighting. Various sensors such as PIR sensors, MQ gas sensor, thermistor and LDR are used to continuously monitor environmental conditions like motion, temperature, smoke presence and ambient light intensity. These sensors provide real-time input data to the microcontroller.

Based on the received inputs and predefined logic, the microcontroller processes the data and controls output devices such as a DC motor (fan) through an L298N motor driver, a servo motor for door operation, LEDs for indication and a buzzer for alert generation. The system also incorporates a security mode, which alters the behaviour of certain modules to enhance protection during critical conditions, such as restricting door access and enabling intrusion alerts.

The modular design of the system ensures that each function operates independently while being coordinated by the central controller. This makes the system flexible, easy to maintain and suitable for further expansion. Overall, the system provides an effective and low-cost solution for smart home automation with real-time monitoring and automated decision-making capabilities.

2.2 BLOCK DIAGRAM EXPLANATION :

The block diagram of the proposed smart home automation system represents the interaction between the central controller, input sensors and output devices. The Arduino Uno microcontroller acts as the core processing unit, which receives input signals from various sensors and controls the corresponding output devices based on programmed logic.

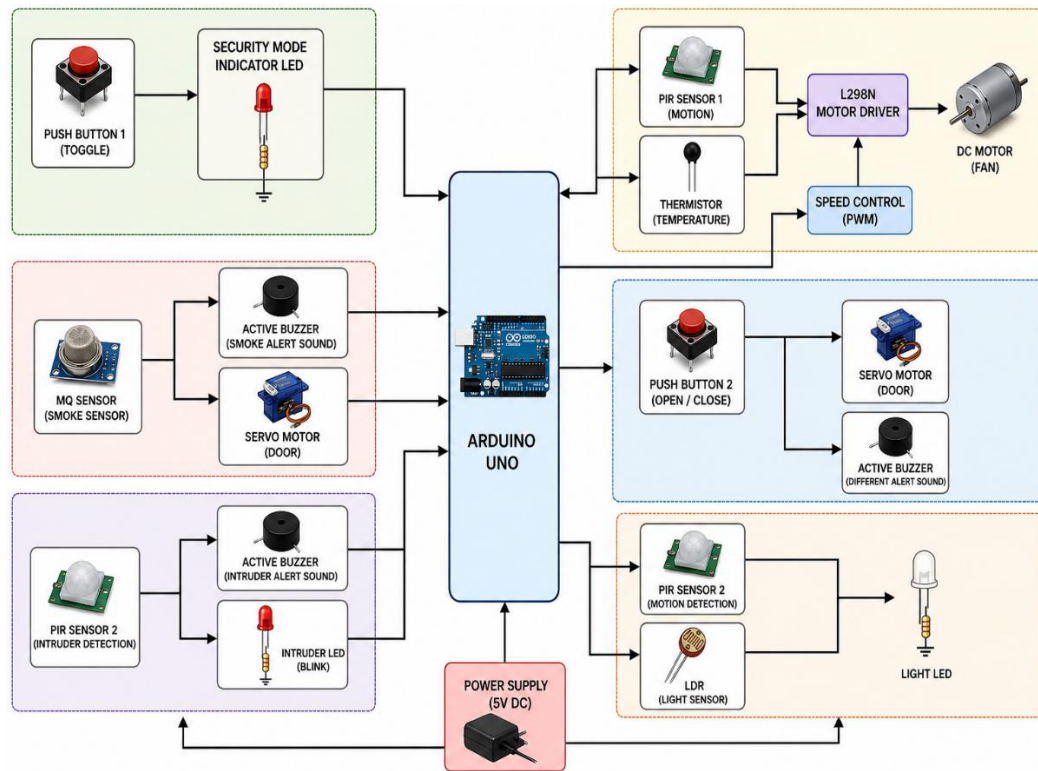


Fig 2.1 Block Diagram of Smart Home Automation System

The input section of the system consists of multiple sensors and user input devices. Two push buttons are used, where one button is dedicated to toggling between normal mode and security mode and the other is used for manual door control. PIR sensors are employed for motion detection; one is used for controlling the fan and lighting system, while the second PIR sensor is used for intruder detection. A thermistor is used to measure temperature, an MQ sensor is used for detecting smoke or gas leakage and an LDR is used to sense ambient light intensity. These sensors continuously provide analog or digital signals to the microcontroller.

The processing unit, which is the Arduino Uno, reads all sensor inputs using its digital and analog input pins. Based on predefined threshold values and logical conditions, it processes the data and determines the appropriate action. The system also includes a security mode feature, which acts as a global control condition and modifies the behaviour of certain modules such as door access and alert mechanisms.

The output section consists of actuators and indicators controlled by the microcontroller. A DC motor, connected through an L298N motor driver module, is used for fan operation, where speed is controlled using PWM signals. A servo motor is used to operate the door by rotating to specific angles for opening and closing. LEDs are used as indicators for security status, intruder detection and automatic lighting. A buzzer is used to generate

different alert sounds for smoke detection, unauthorized access and intrusion alerts.

The power supply unit provides the required DC voltage to all components in the system. The microcontroller coordinates all modules, ensuring proper communication between input and output devices. Overall, the block diagram illustrates a centralized control system where real-time sensor data is processed to perform automated actions, ensuring safety, energy efficiency and user convenience.

2.3 HARDWARE COMPONENTS DESCRIPTION:

- **Arduino Uno:** The Arduino Uno is the central microcontroller used to control the entire system. It processes inputs from sensors and generates appropriate outputs based on programmed logic. It supports both analog and digital I/O operations, thus making it suitable for interfacing multiple components.



Fig 2.2 Arduino Uno

- **PIR Sensor (Passive Infrared Sensor) :** The PIR sensor is used to detect motion by sensing infrared radiation emitted by human bodies. In this project, one PIR sensor is used for fan and lighting control, while another is used for intruder detection. It provides a digital HIGH signal when motion is detected.



Fig 2.3 PIR Sensor

- **MQ Gas Sensor:** The MQ sensor is used to detect the presence of smoke or gas in the

environment. It provides an analog output proportional to gas concentration. When the detected value exceeds a predefined threshold, the system triggers safety actions such as activating the buzzer.



Fig 2.4 MQ Gas Sensor

- **Thermistor:** The thermistor is a temperature-sensitive resistor used to measure ambient temperature. Its resistance varies with temperature and this variation is converted into an analog voltage read by the microcontroller. It is used to control the fan speed based on temperature levels.



Fig 2.5 Thermistor

- **LDR (Light Dependent Resistor):** The LDR is used to measure light intensity. Its resistance decreases as light intensity increases. The microcontroller reads its analog value to determine whether the environment is dark or bright, enabling automatic lighting control.

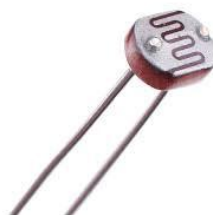


Fig 2.6 LDR (Light Dependent Resistor)

- **L298N Motor Driver Module :** The L298N motor driver is used to control the DC motor

(fan). It allows the microcontroller to control motor speed and direction using PWM signals. It also provides sufficient current to drive the motor safely.

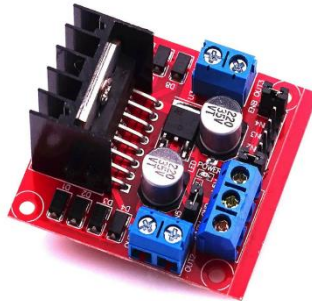


Fig 2.7 L298N Motor Driver Module

- **DC Motor** : The DC motor is used as a fan in this system. Its speed is controlled using PWM signals through the motor driver. The motor operates at different speeds based on temperature conditions.



Fig 2.8 DC Motor

- **Servo Motor** : The servo motor is used for door control. It rotates to specific angles (e.g., 0° for closed and 90° for open) based on control signals from the microcontroller. It provides precise position control required for door operation.



Fig 2.9 Servo Motor

- **Buzzer** : The buzzer is used as an alert device. It generates sound signals for different

conditions such as smoke detection, intruder alert and unauthorized access. Different frequencies are used to distinguish between alerts.



Fig 2.10 Buzzer

- **Push Buttons :** Push buttons are used for user input. One button is used to toggle between normal and security modes, while the other is used for manual door control. They provide digital input signals to the microcontroller.



Fig 2.11 Push Button

- **LEDs :** LEDs are used as visual indicators in the system. A security LED indicates the current mode, an intruder LED signals motion detection and a light LED is used for automatic lighting.



Fig2.12 LED

2.4 SOFTWARE DESCRIPTION :

The software for the proposed smart home automation system is developed using the Arduino IDE and is written in Embedded C. The program is designed to continuously monitor sensor inputs and control output devices based on predefined conditions. The system operates in a loop, ensuring real-time response to environmental changes.

I. Initialization:

In the setup phase, all input and output pins are configured using appropriate modes:

- Push buttons are configured as INPUT_PULLUP
- PIR sensors are set as digital inputs
- Analog sensors (thermistor, MQ sensor, LDR) are read through analog input pins
- Output devices such as motor driver, buzzer, LEDs and servo motor are initialized

The servo motor is set to the default closed position (0°) at startup. Serial communication is also initialized for debugging and monitoring sensor values.

II. Mode Selection Logic

The system includes a **security mode toggle mechanism**:

- A push button is used to switch between **normal mode** and **security mode**
- Each button press toggles the system state
- An LED indicates the current mode:
 - LED OFF → Normal Mode
 - LED ON → Security Mode

This mode selection affects the behaviour of certain modules like door control and alert systems.

III. Sensor Data Acquisition

The microcontroller continuously reads input data from all sensors:

- **Thermistor** → Temperature value (analog)
- **MQ Sensor** → Smoke/gas level (analog)
- **LDR** → Light intensity (analog)
- **PIR Sensors** → Motion detection (digital)

These values are compared with threshold levels to determine system actions.

IV. Temperature and Fan Control Logic:

- The system checks for motion using PIR1
- If motion is detected:
 - Temperature values are evaluated against three thresholds
 - Motor speed is adjusted using PWM signals:
 - Low temperature → Low speed
 - Medium temperature → Moderate speed
 - High temperature → Maximum speed
- If no motion is detected, the motor remains OFF

V. Smoke Detection and Safety Logic:

- The MQ sensor continuously monitors smoke levels
- If the value exceeds a threshold:
 - The buzzer is activated to alert users
- System behaviour depends on mode:
 - **Normal Mode:** Door opens automatically using servo motor
 - **Security Mode:** Door remains closed, only buzzer alerts

VI. Door Control Logic

- A push button is used to manually control the door
- In **normal mode:**
 - Each button press toggles the door state (open/close)
- In **security mode:**
 - Door operation is disabled
 - A warning sound is generated using the buzzer

VII. Intruder Detection Logic :

- PIR2 detects motion for intrusion monitoring
- In **normal mode:**
 - An LED blinks to indicate motion
- In **security mode:**
 - The buzzer is activated with a distinct tone to indicate intrusion

VIII. Automatic Lighting Logic:

- The system uses **PIR1 and LDR**
- If motion is detected and light intensity is below a threshold:
 - The LED light is turned ON
- Otherwise:
 - The light remains OFF

IX. Alert Priority Handling:

- The system prioritizes **intruder alerts over other alerts**
- Different buzzer frequencies are used for:
 - Smoke detection
 - Unauthorized door access
 - Intrusion detection

This ensures clear differentiation of events.

X. Continuous Loop Operation:

All modules operate within the main loop, ensuring:

- Continuous monitoring of sensors
- Immediate response to environmental changes
- Real-time automation and control

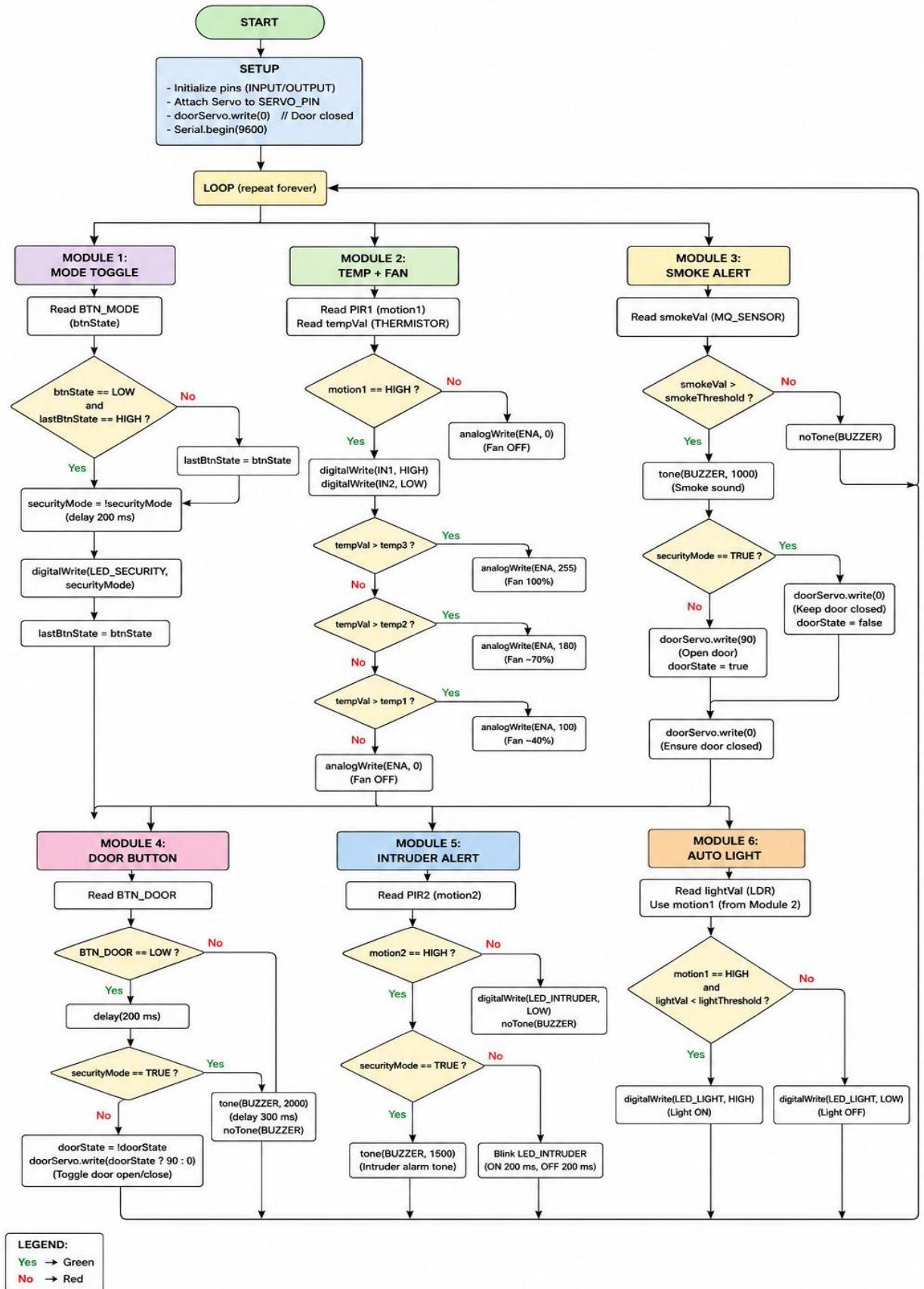


Fig 2.13 Software Flowchart of Smart Home Automation System

SOURCE CODE:

```

#include <Servo.h>

Servo doorServo;

// ----- PIN DEFINITIONS -----

#define BTN_MODE 5
#define BTN_DOOR 13
#define PIR1 2//for light and fan
#define PIR2 8// for intruder alarm
#define BUZZER 3
#define IN1 11
#define IN2 12
#define ENA 10
#define SERVO_PIN 9
#define LED_SECURITY 6
#define LED_INTRUDER 7
#define LED_LIGHT 4

#define THERMISTOR A1
#define MQ_SENSOR A2
#define LDR A0

// ----- VARIABLES -----

bool securityMode = false;
bool lastBtnState = HIGH;
bool doorState = false; // false = closed, true = open

// threshold values
int temp1 = 100;

```

```

int temp2 = 200;
int temp3 = 300;
int smokeThreshold = 150;
int lightThreshold = 100;

// ----- SETUP -----
void setup() {
  pinMode(BTN_MODE, INPUT_PULLUP);
  pinMode(BTN_DOOR, INPUT_PULLUP);
  pinMode(PIR1, INPUT);
  pinMode(PIR2, INPUT);
  pinMode(BUZZER, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(ENA, OUTPUT);
  pinMode(LED_SECURITY, OUTPUT);
  pinMode(LED_INTRUDER, OUTPUT);
  pinMode(LED_LIGHT, OUTPUT);
  doorServo.attach(SERVO_PIN);
  doorServo.write(0); // door closed

  Serial.begin(9600);
}

// ----- LOOP -----
void loop() {

  // ----- MODULE 1: MODE TOGGLE -----
  bool btnState = digitalRead(BTN_MODE);

```

```

if (btnState == LOW && lastBtnState == HIGH) {
    securityMode = !securityMode;
    delay(200); // debounce
}
lastBtnState = btnState;
digitalWrite(LED_SECURITY, securityMode);

// Read sensors
int tempVal = analogRead(THERMISTOR);
int smokeVal = analogRead(MQ_SENSOR);
Serial.println(tempVal);
int lightVal = analogRead(LDR);

bool motion1 = digitalRead(PIR1);
Serial.println(motion1);
bool motion2 = digitalRead(PIR2);

// ----- MODULE 2: TEMP + FAN -----
if (motion1) {
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);

    if ((tempVal > temp1) && (tempVal < temp2)) {
        analogWrite(ENA, 100);
    }
    else if ((tempVal > temp2) && (tempVal < temp3)) {
        analogWrite(ENA, 180);
    }
    else if (tempVal > temp3) {

```

```

    analogWrite(ENA, 255);
}
else {
    analogWrite(ENA, 0);
}
} else {
    analogWrite(ENA, 0);
}

// ----- MODULE 3: SMOKE ALERT -----
if (smokeVal > smokeThreshold) {
    tone(BUZZER, 1000); // smoke sound

    if (!securityMode) {
        doorServo.write(90); // open
        doorState = true;

    } else {
        doorServo.write(0); // keep closed
        doorState = false;
    } doorServo.write(0);
} else {
    noTone(BUZZER);
}

// ----- MODULE 4: DOOR BUTTON -----
if (digitalRead(BTN_DOOR) == LOW) {
    delay(200);
}

```

```

if (!securityMode) {
    doorState = !doorState;
    doorServo.write(doorState ? 90 : 0);
} else {
    tone(BUZZER, 2000); // different alert
    delay(300);
    noTone(BUZZER);
}
}

// ----- MODULE 5: INTRUDER ALERT -----
if (motion2) {
    if (!securityMode) {
        digitalWrite(LED_INTRUDER, HIGH);
        delay(200);
        digitalWrite(LED_INTRUDER, LOW);
    } else {
        tone(BUZZER, 1500); // different tone
    }
} else {
    digitalWrite(LED_INTRUDER, LOW);
}

// ----- MODULE 6: AUTO LIGHT -----
if (motion1 && lightVal < lightThreshold) {
    digitalWrite(LED_LIGHT, HIGH);
} else {
    digitalWrite(LED_LIGHT, LOW);
}
}

```


2.5 CIRCUIT DESCRIPTION:

The circuit of the proposed smart home automation system is centered around the Arduino Uno microcontroller, which interfaces with various input sensors and output devices through its digital and analog pins. Proper power distribution and grounding are maintained to ensure stable operation of all components.

I. Power Supply Connections:

- The Arduino Uno is powered via USB or an external DC supply.
- The L298N motor driver is powered separately using a higher voltage battery (e.g., 8–12V).
- All components share a **common ground (GND)** to ensure proper signal reference.

II. Input Device Connections:

Push Buttons:

- **Mode Toggle Button (BTN_MODE)** → Connected to **Digital Pin 5**
- **Door Control Button (BTN_DOOR)** → Connected to **Digital Pin 13**
- Configured using **INPUT_PULLUP**, so one terminal goes to GND.

PIR Sensors

- **PIR1 (Fan + Light control)** → Digital Pin 2
- **PIR2 (Intruder detection)** → Digital Pin 8
- Output pin of PIR → Arduino digital input
- VCC → 5V, GND → GND

Analog Sensors

- **Thermistor** → Analog Pin A1 (via voltage divider)
- **MQ Gas Sensor** → Analog Pin A2
- **LDR** → Analog Pin A0 (via voltage divider)

These sensors provide analog voltage signals to the Arduino.

III. Output Device Connections:

Motor (Fan) with L298N Driver

- **IN1** → **Pin 11**
- **IN2** → **Pin 12**
- **ENA** → **Pin 10 (PWM pin)**
- Motor is connected to output terminals of L298N

- PWM signal controls speed based on temperature

Servo Motor (Door Control)

- Signal pin → **Digital Pin 9**
- VCC → 5V
- GND → GND

Controls door position (0° closed, 90° open)

Buzzer

- Positive → **Digital Pin 3**
- Negative → GND

Generates different tones using tone() function

LED Indicators

- **Security Mode LED → Pin 6**
- **Intruder LED → Pin 7**
- **Automatic Light LED → Pin 4**

Each LED is connected with a current-limiting resistor ($\sim 220\Omega$)

IV. Grounding and Signal Integrity:

- All modules share a **common ground** with Arduino
- Proper grounding ensures accurate sensor readings and stable actuator operation
- Analog sensors use **voltage divider circuits** for proper signal scaling

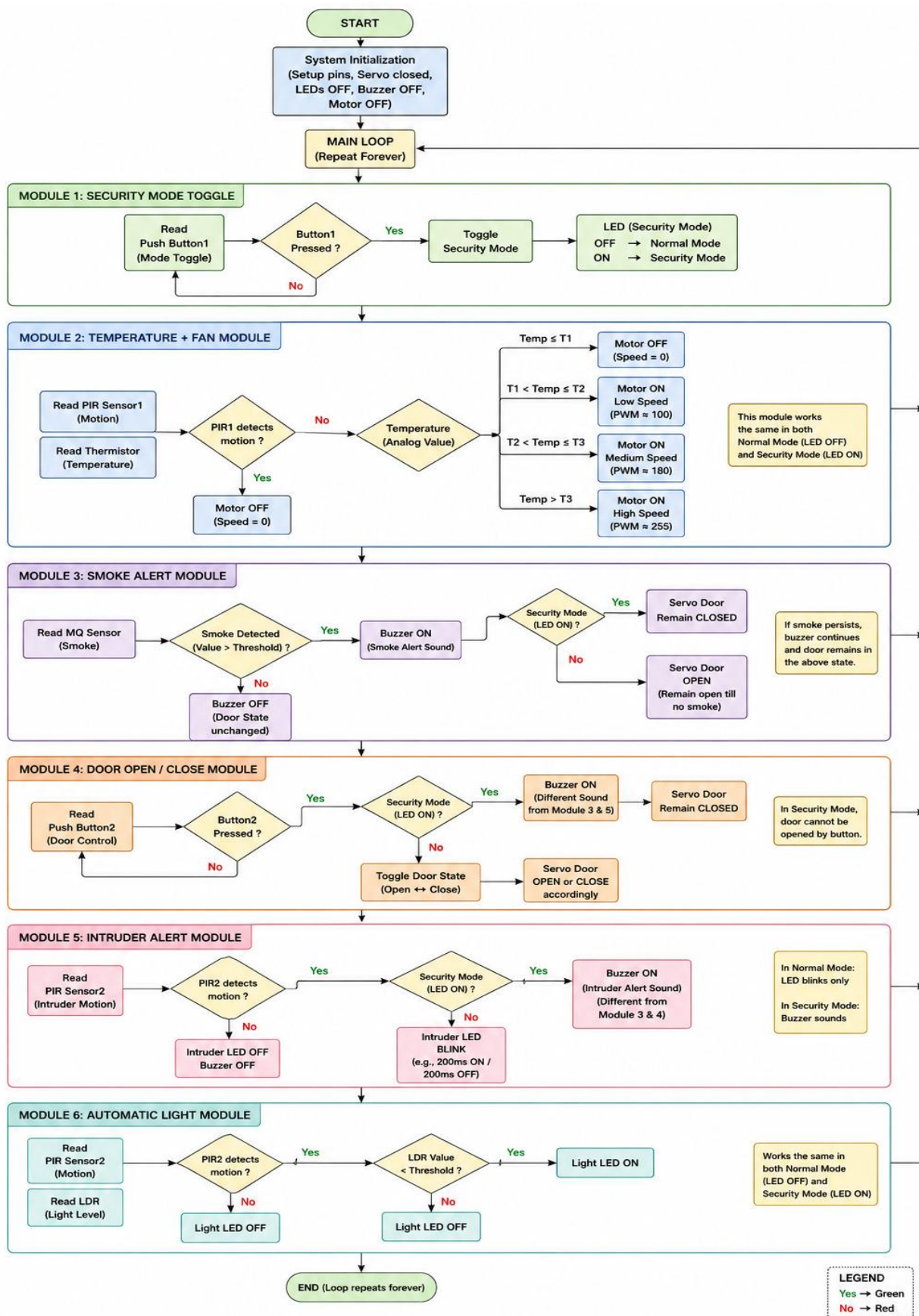


Fig 2.14 System Architecture of Smart Home Automation System

CHAPTER 3

RESULT AND DISCUSSION

3.1 SYSTEM WORKING:

The proposed smart home automation system operates by continuously monitoring environmental conditions through sensors and controlling various devices based on predefined logic programmed into the microcontroller. The Arduino Uno acts as the central control unit, coordinating all modules in real time.

Initially, the system is powered ON and enters the default normal mode, where the security LED remains OFF. The user can toggle between normal mode and security mode using a push button. When the button is pressed, the system switches its mode and the corresponding LED indicates the current state.

The system continuously reads input data from sensors including PIR sensors, thermistor, MQ gas sensor and LDR. Based on these inputs, the microcontroller performs specific actions:

- When motion is detected by PIR1 and the temperature exceeds predefined threshold levels, the DC motor (fan) is activated. The motor speed is adjusted using PWM signals according to the temperature intensity.
- When the MQ sensor detects smoke or gas above the threshold value, the buzzer is activated to alert users. In normal mode, the servo motor automatically opens the door and keeps it open until the smoke level reduces. In security mode, the door remains closed while the buzzer continues to alert.
- The door can be manually controlled using a push button. In normal mode, each press toggles the door between open and closed states. In security mode, manual door operation is disabled, and an alert sound is generated.
- Intruder detection is performed using PIR2. In normal mode, motion detection causes an LED to blink. In security mode, a distinct buzzer alert is generated to indicate unauthorized presence.
- The automatic lighting system uses PIR1 and LDR. When motion is detected and the ambient light is low, the LED light turns ON automatically, ensuring efficient energy usage.

The system continuously executes these operations in a loop, ensuring real-time responsiveness and automation.

3.2 OBSERVATION:

During the implementation and testing of the system, the following observations were made:

- The system successfully toggles between normal mode and security mode using the push button, with clear indication through the LED.
- The temperature-based fan control operates effectively with noticeable changes in motor speed corresponding to different temperature levels.
- The smoke detection module responds promptly when gas or smoke is detected, triggering the buzzer and appropriate door operation based on the selected mode.
- The door control mechanism functions reliably, allowing manual operation in normal mode and restricting access in security mode.
- The intruder detection module accurately detects motion and provides different responses based on the system mode.
- The automatic lighting system works efficiently by turning ON the light only under low-light and motion conditions, thereby conserving energy.
- The buzzer produces distinct sounds for different alerts, making it easy to differentiate between smoke detection, intrusion, and unauthorized access.
- Overall, the system demonstrates stable performance, quick response to sensor inputs and effective coordination between multiple modules.

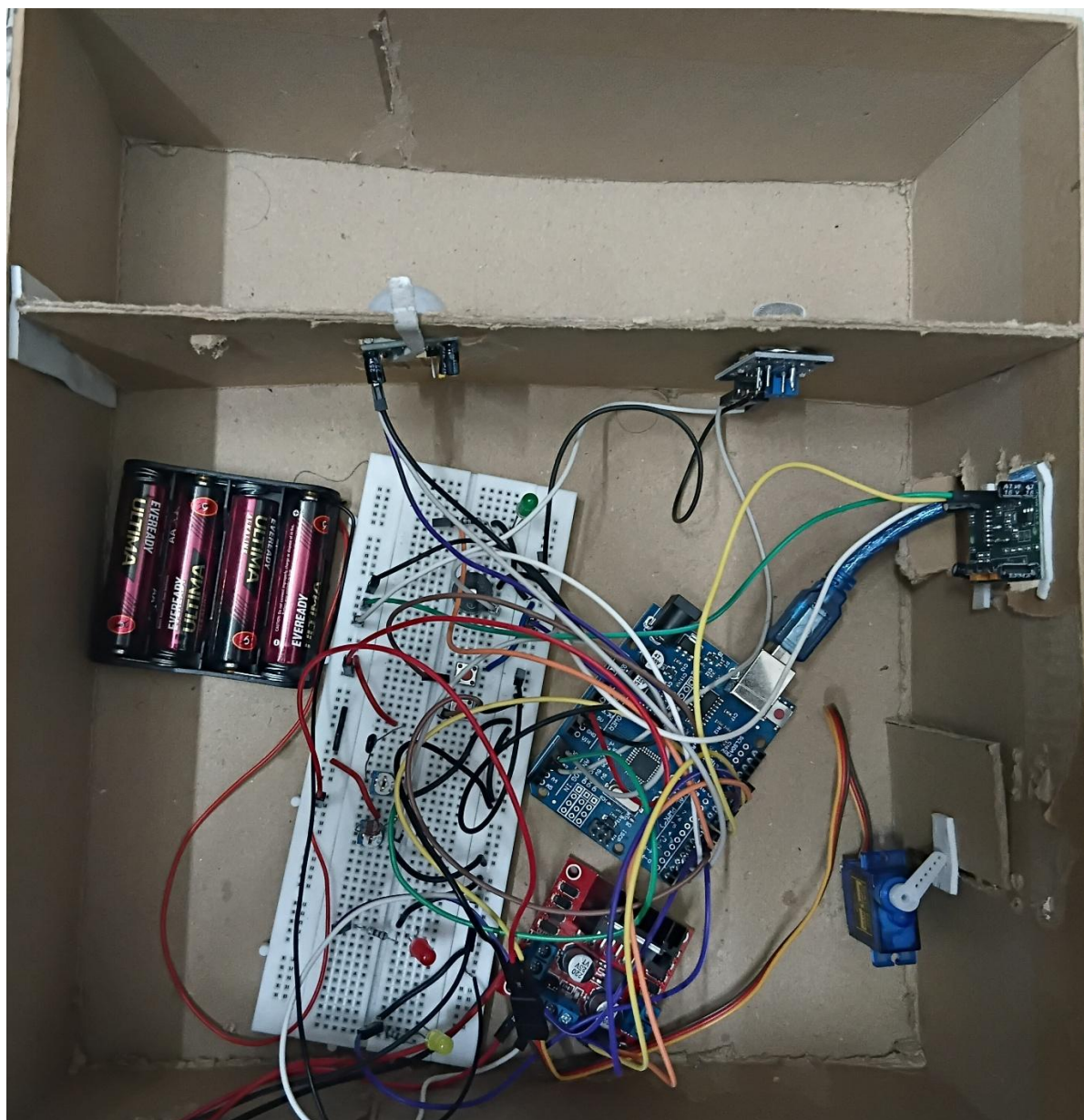


Fig 3.1 Final Output

3.3 FINAL RESULT AND OVERALL PERFORMANCE:

The proposed multi-module smart home automation system was successfully designed, implemented and tested. All six modules; security mode control, temperature-based fan control, smoke detection, door control, intruder alert and automatic lighting operated as intended according to the programmed logic.

The system effectively demonstrated real-time monitoring and control of environmental conditions using various sensors. The temperature and motion-based fan control functioned accurately with smooth variation in motor speed achieved through PWM. The smoke detection module responded promptly to gas presence, activating alerts and performing appropriate door control based on the system mode.

The security mode feature proved to be reliable, successfully altering the behavior of critical modules such as door access and intrusion alerts. The intruder detection system accurately identified motion and generated distinct alerts, while the automatic lighting module ensured efficient operation by activating lights only when required.

The integration of multiple modules into a single system was achieved without conflicts and the use of different alert tones helped in clearly distinguishing between various system events. The system showed stable performance during continuous operation with consistent response to sensor inputs and minimal delay in execution.

Overall, the project achieved its intended objectives by providing a low-cost, reliable and efficient smart home automation solution. The system demonstrates good scalability, modularity and practical applicability in real-world residential environments, thus making it a suitable foundation for further enhancements and IoT integration.

CHAPTER-4

CONCLUSION

The proposed smart home automation system was successfully designed and implemented using a microcontroller-based approach. The system integrates multiple functional modules, including security mode control, temperature-based fan operation, smoke detection, door control, intruder alert and automatic lighting to provide a comprehensive and efficient home automation solution.

The project achieved its primary objective of enhancing safety, convenience and energy efficiency in a residential environment. The system was able to monitor real-time environmental conditions using various sensors and respond effectively through automated control of actuators. The implementation of a security mode further improved system reliability by restricting unauthorized access and enabling prioritized alert mechanisms.

The modular design of the system allowed seamless integration of all components, ensuring stable and coordinated operation without conflicts. The use of PWM-based motor control, distinct alert signals and condition-based logic contributed to the overall effectiveness and responsiveness of the system.

In conclusion, the developed system demonstrates a low-cost, reliable and scalable approach to smart home automation. It provides a strong foundation for future enhancements and highlights the potential of embedded systems in creating intelligent and user-friendly living environments.

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